

PCT/PCM 9: How Far Can One Neuron See?

4. Interpret the Figure

For plain 3×3 stacking with stride 1, the receptive field grows linearly because each layer adds only 2 pixels. When stride-2 layers are inserted, the curve bends upward and grows much faster because later kernels operate with a larger jump on the original input grid.

5. Receptive Field Calculation

For stride-1 3×3 convolutions, each layer adds $(3-1) = 2$ pixels.

Start: $RF = 1$

After 1 layer: $RF = 3$

After 2 layers: $RF = 5$

After 3 layers: $RF = 7$

General rule for k stacked 3×3 stride-1 layers:

$$RF = 1 + 2k.$$

6. Prediction

For a 200-pixel-wide object:

$$1 + 2k \geq 200$$

$$2k \geq 199$$

$$k \geq 99.5$$

Therefore, about 100 plain 3×3 stride-1 layers would be needed. This is impractical, which is why stride and dilation are useful for increasing receptive-field size more efficiently.